Tristan Ko

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Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science (BASc), Management Engineering Sep 2024 - Apr 2029

- Relevant Coursework: Data Structures & Algorithms, Probability & Statistics, Statistical Modeling, Optimization

Experience

Provincial Product Analyst Co-op Sep 2026 - Dec 2026
Allstate Insurance Company of Canada *Markham, ON*

- Pinpointed the least profitable segments of the tenant multiline and auto portfolios and used them to shape a new **ceding strategy** for the risk-sharing pool, independently surfacing an area that became a **targeted focus** for the business
- Automated monthly data pulls and analysis with **Python (Pandas)**, cutting report prep time by **85%** and keeping **12+** recurring reports accurate and on schedule for stakeholders
- Supported the team's migration from **SAS** to **Microsoft Fabric** while maintaining **Power BI** dashboards that give product managers a clear view of provincial performance

Junior Technical Analyst Jan 2026 - Apr 2026
Ontario Ministry of Transportation *Toronto, ON*

- Diagnosed and resolved critical data-flow issues between an **Angular** frontend, a **Spring Boot API**, and an **Azure SQL** database by tracing application caching logic and correcting database lifecycle timelines
- Engineered **idempotent SQL** migration scripts to refactor multi-row operations, successfully managing Unique Index constraints during production-level data refreshes to ensure **100%** data consistency
- Developed a centralized **SQL** Server database to manage Alternate Work Arrangement (AWA) tracking, migrating the ministry from manual spreadsheets to a scalable, organized data repository

Supply Chain Intern May 2025 - Aug 2025
Iovate Health Sciences International Inc. *Oakville, ON*

- Developed an **Excel** buying tool to calculate reorder timing and quantities across **800+** active SKUs, aligning purchases with Days of Supply targets to reduce manual planning and decision time by **90%+**
- Built a Python script with **Pandas** to automate **ERP** uploads, eliminating manual entry and data entry errors
- Designed a **Power BI** dashboard centralizing key supply metrics, cutting report search time and improving visibility for the planning team
- Analyzed warehouse deviation data using **Excel** to identify systemic issues, reducing fulfillment errors by **35%**

Projects

Code Ranch | *TypeScript, Next.js, React, Tailwind CSS, PostgreSQL* Live

- Developed a gamified syntax typing game for programmers using **Next.js** and **TypeScript** with multiple modes and stats
- Implemented **real-time multiplayer** duels using Supabase Realtime to sync code and live progress with **<100ms** latency
- Designed a **normalized PostgreSQL** schema and implemented Supabase **RPC** to manage relational data for social systems, online presence, and user metrics

Factor Five | *TypeScript, Next.js, React, Node.js, Tailwind CSS* Live

- Engineered a 5-factor **scoring engine** using robust statistics to benchmark stocks across size-matched peer cohorts
- Built a **dual-priority** request pipeline with single-flight deduplication to enforce strict **60 req/min API** limits
- Optimized Next.js performance via a **165KB** client search index, enabling **sub-2ms** lookup across **5,350+** US equities

Stock Market Classification Model | *Python, Pandas, NumPy, Scikit-learn, XGBoost, yfinance* GitHub

- Built classification models for SPY, QQQ, DJI & IWM to predict 1D, 5D and 20D price direction with ∼**53-57%** accuracy
- **Automated** an **ETL** pipeline using yfinance to ingest data, update features, and generate daily price predictions
- Validated model performance using time-based cross-validation and **F1-score** to ensure stability across market regimes

Skills

Languages: Python, SQL, JavaScript/TypeScript, Java, R, C/C++, VBA
Libraries: Pandas, NumPy, Scikit-learn, XGBoost, Gurobi (gurobipy), Matplotlib, Plotly
Frameworks: Spring Boot, FastAPI, Angular, React